

AI-Powered Crowd Detection Using Wi-Fi Access Points (WPS) with User Interface for University Campus

Problem Statement:

Managing campus crowd flow through CCTV or manual counting is inefficient, costly, and raises privacy concerns. This project proposes an **AI-driven system that estimates and visualizes crowd density using Wi-Fi Access Point (WPS) signals**, offering real-time, anonymous monitoring for campus safety and resource planning.

Motivation / Introduction:

Every student carries Wi-Fi-enabled devices that continuously emit signals. By analyzing these signals—especially **RSSI values and probe requests**—we can estimate people density without tracking identities. The goal is to design a **smart campus platform** that provides live crowd heatmaps, helping manage classrooms, cafeterias, and events efficiently while preserving privacy.

Dataset:

Data will be collected via existing Wi-Fi APs or Raspberry Pi sniffers in monitor mode. Each record includes **timestamp, signal strength (RSSI), AP ID, and anonymized device identifiers (hashed MACs)**. Ground-truth counts will be obtained manually to train and validate models.

Literature Review:

Previous studies like *Faruqi et al. (2020)* and *Amin et al. (2019)* demonstrated over **85–90% accuracy** in people counting using Wi-Fi probe data. Recent work also integrates **machine learning models (Random Forest, Neural Networks)** for enhanced precision. However, few provide real-time visualization. This project bridges that gap by combining **AI-based estimation with an interactive dashboard** for practical campus deployment.

Tasks / Methodology:

We will collect Wi-Fi data, clean and preprocess it, extract features (like average RSSI), and train models to estimate crowd counts. A backend (FastAPI) will process data, and a frontend (React + Tailwind) will display real-time heatmaps. The system's accuracy will be tested using ground-truth comparisons and metrics such as MAE and RMSE.

Scope:

Real-time, privacy-safe crowd detection using Wi-Fi with an AI model, live dashboard, and accuracy evaluation.